Ragnarok: Trial of the Nine Realms

Dashboard Guide

What every part of the Ragnarok website does, how to run it, and how to use it to investigate a supply-chain disruption from start to finished briefing.

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Screenshots are from the saved two-run case `case-20261025-204050-ac35da` on the dev pack (as of 25 January 2027). Everything the dashboard shows comes from the case's recorded events and the API. Nothing on screen is simulated.

1. What the website is for

Ragnarok answers questions like *"As of 25 January 2027, how exposed are Midgard and Tolan to the end of the Orvan corridor agreement? Give a low and high range for wheat and maize imports."* It does this with four AI agents that work over the Runemark document feed:

AGENT	ROLE	WHERE YOU SEE IT
Heimdall	Investigates first. Searches the documents and trade tables, checks the event is real, notes corrections and rumours, lists missing data and builds the **Case file**. If Odin sends the case back, Heimdall runs new targeted searches.	Trial (Case file), Run trace (send-back research)
Thor	Argues for **high** exposure, with cited claims and a high estimate.	Trial (left column), Exposure
Loki	Argues independently for **limited** exposure (substitutes, other routes, stocks), with a low estimate.	Trial (right column), Exposure
Odin	Judges. Rules on every claim (established, contested, conceded, rejected), builds his own range from evidence, may send the case back for more evidence, and writes the final briefing from his rulings only.	Trial (rulings), Exposure, Briefing, Run trace

The flow is fixed in code: **Question → Heimdall → Thor and Loki (in parallel) → Cross-examination → Odin → Briefing**, or, if Odin finds essential evidence missing, **Heimdall's new research → second run**. The website lets you start a case, watch it run live, inspect every claim and citation, and export the final briefing.

2. Starting the website

Two processes must run: the backend API (port 8000) and the dashboard (port 5173). From the project folder:

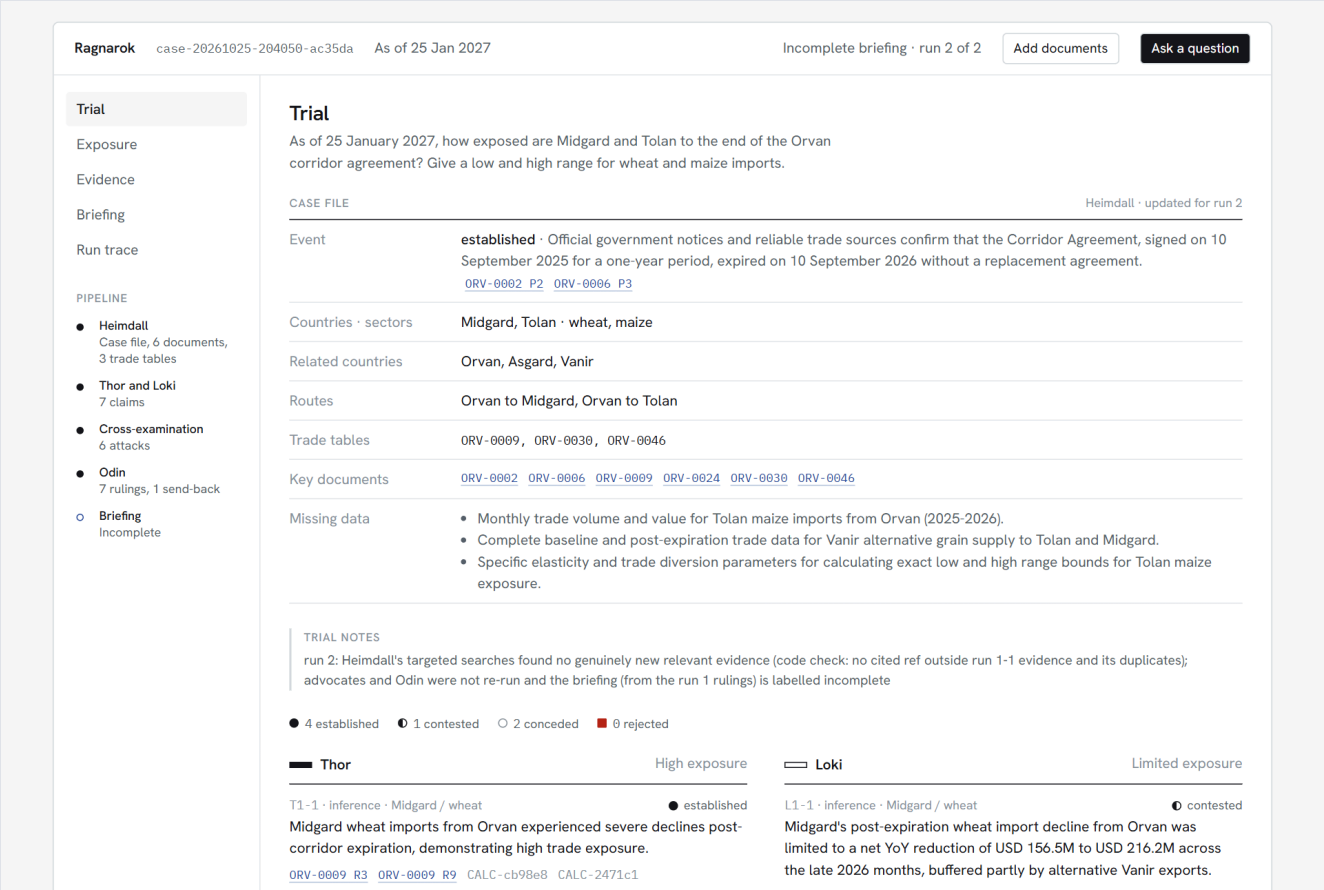
```
# 1) API, from the project root
.venv/Scripts/python -m uvicorn ragnarok.api:app --app-dir backend --port 8000

# 2) Dashboard, from frontend/
npm run dev          # then open http://localhost:5173
```

The dashboard forwards its requests to the API on port 8000. The Gemini API key lives in `.env` (never on the website). Interactive API documentation is at `http://localhost:8000/api/docs`.

> **Links you can share.** The address bar holds the case and the view, for example `http://localhost:5173/#case=case-20261025-204050-ac35da&view=exposure`. Opening that link goes straight to that case and view. The browser Back button moves between views.

3. Screen layout



A case open on the Trial view. Top bar across the top, side rail on the left, the selected view on the right.

Top bar

ITEM	WHAT IT DOES
Ragnarok · case ID	The ID of the open case (for example `case-20261025-204050-ac35da`). Copy it to reopen the case later.
As of <date>	The case's as-of date. Nothing published after it is ever shown to the agents or to you.
Run status	Live status text, for example *Running · Odin · run 1 of 2*, *Paused · daily model quota*, *Complete briefing · run 2 of 2* or, in red, *Failed at Heimdall · no briefing*. See section 13.
Add documents	Opens the upload dialog: add a new event pack or single documents to the index (section 5).
Ask a question	Opens the question dialog: start a new investigation or open a saved case (section 5).

Side rail

Views: Trial, Argument map, Exposure, Evidence, Briefing, Run trace. Click to switch; the open case stays the same.

Pipeline: the five trial stages with a live one-line summary each (documents and trade tables found, number of claims, attacks, rulings and send-backs, briefing status). The dot shows the state of each stage:

● done ○ running now (or the briefing is partial/incomplete) ○ not reached ■ failed here

On a phone the rail becomes a row of view buttons above the page, and the pipeline is hidden.

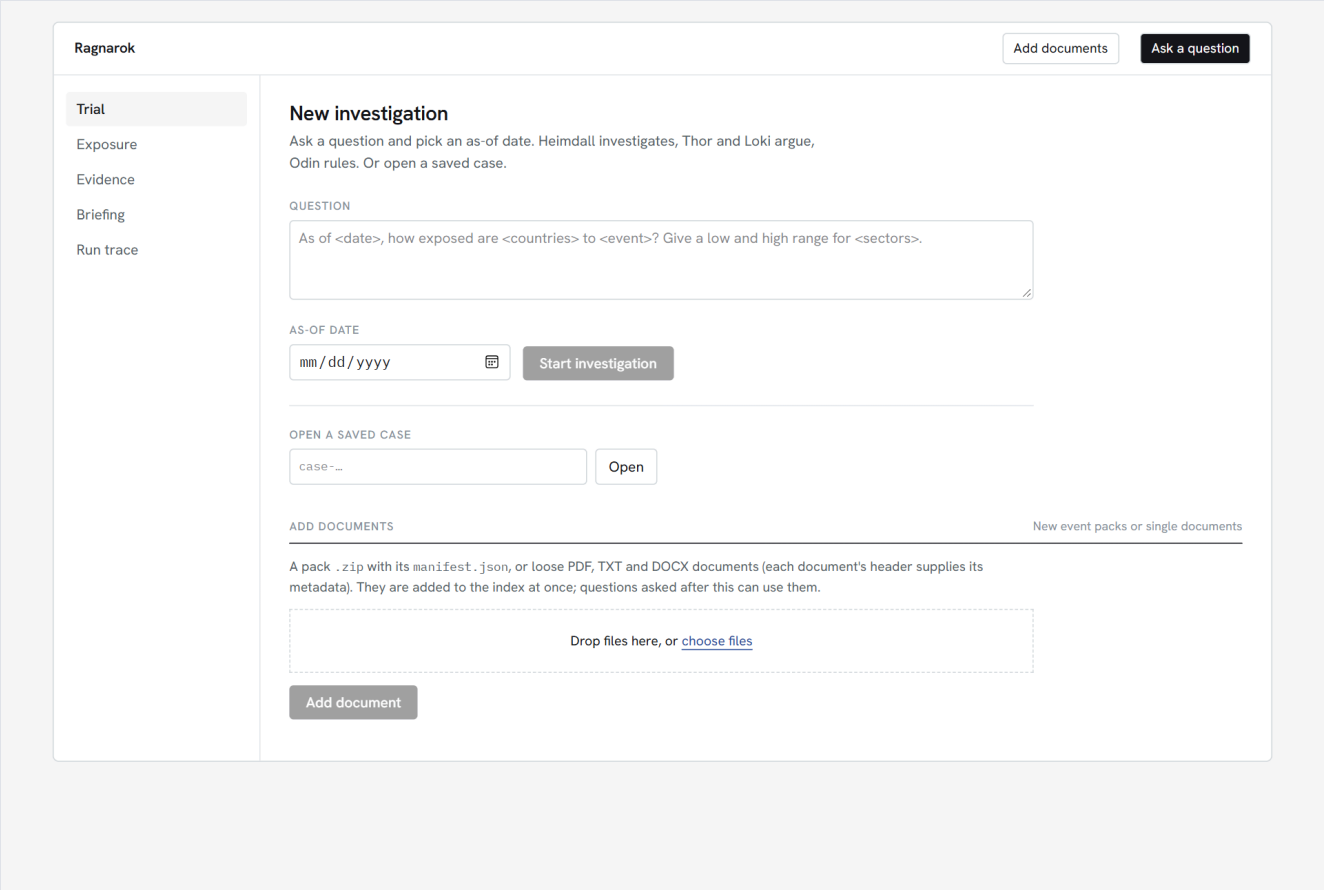
4. Quick start: your first investigation

1. **Open the website** at `http://localhost:5173`. With no case open you see the *New investigation* page.
2. **Type your question.** Name the countries, the event and the sectors, and ask for a low and high range. Example:
   As of 25 January 2027, how exposed are Midgard and Tolan to the end of the Orvan corridor agreement? Give a low and high range for wheat and maize imports.
3. **Pick the as-of date** with the date picker. The agents may only use documents published on or before this date.
4. **Click Start investigation.** The site opens the new case on the Trial view and the case ID appears in the top bar. Only one case runs at a time.
5. **Watch it run.** The pipeline in the side rail updates as each stage finishes. The Evidence view's *Live feed* lists documents as the agents retrieve them. Claims, attacks and rulings appear on the Trial view as they are recorded. A full two-run case takes a few minutes.
6. **Read the result.** Trial shows the debate and Odin's rulings, Exposure shows the three ranges side by side, Briefing shows the final report.
7. **Check any citation.** Click any blue reference such as `ORV-0002 P2`; the Evidence view opens that document with the cited paragraph highlighted.
8. **Export.** On the Briefing view, click `Save as PDF`, `Export .md` or `Export .json`.

> **Reopening a case later:** Ask a question → *Open a saved case* → paste the case ID → `Open` (or press Enter). Saved demo cases live in `logs/demo/`.

> **If the daily model quota runs out** mid-case, the case **pauses** instead of failing. Nothing is lost; it resumes on its own after the quota resets (while the API is running), and the finished steps replay without new requests. See section 13.

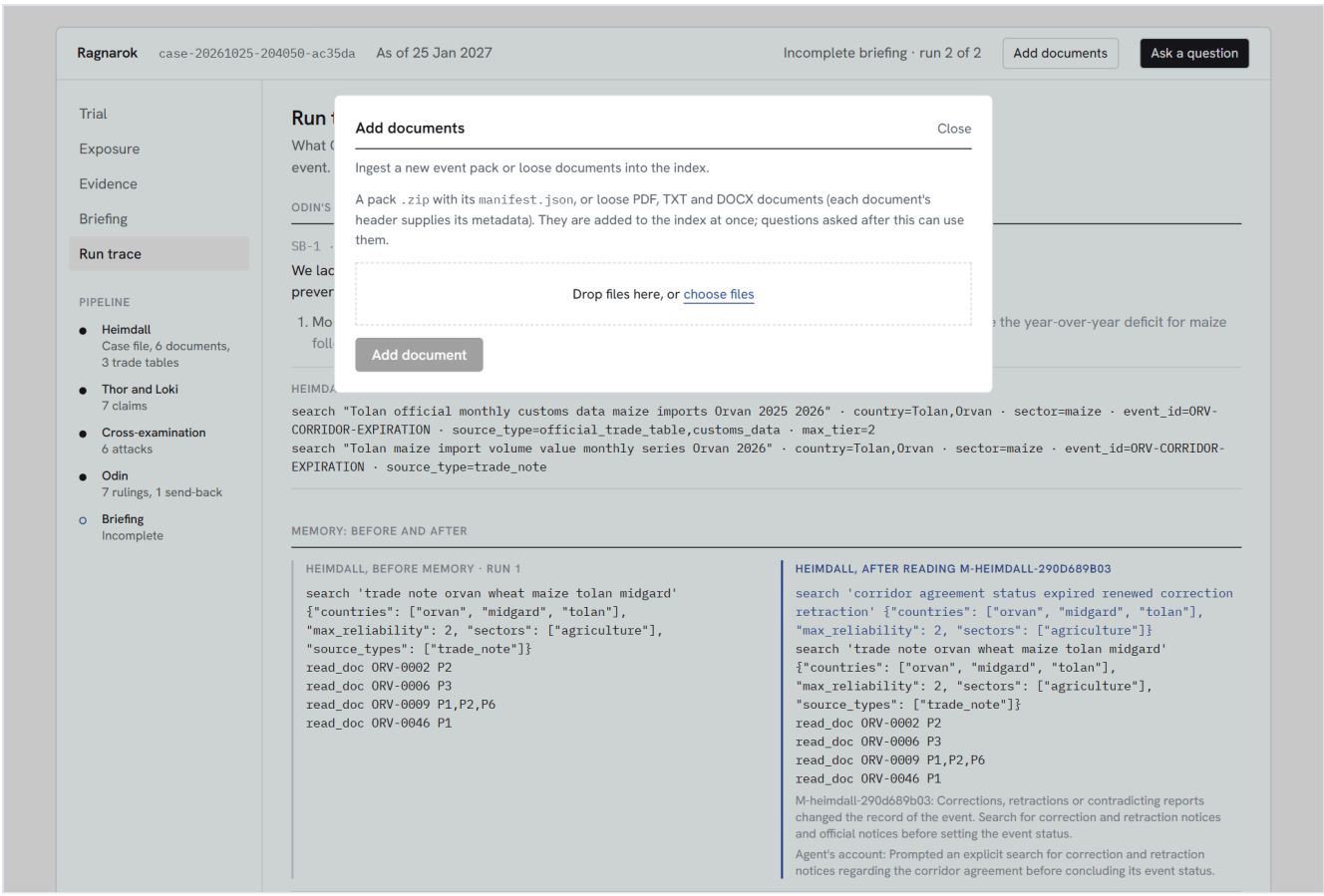
5. Start page and the two dialogs



The start page (no case open): question form, open a saved case, and add documents.

Question form (also in the **Ask a question** dialog)

FIELD / BUTTON	USE
Question	Free text, at least 5 characters. The placeholder shows the expected shape.
As-of date	Required. Sets the evidence cut-off for the whole case.
Start investigation	Creates the case (`POST /cases`) and opens it. Disabled while another case is running ("A case is running; one case at a time.").
Open a saved case	Type or paste a case ID (`case-…`) and click Open.



The *Add documents* dialog.

Add documents (upload)

Adds new evidence to the search index while the site is running; no code changes or restart are needed.

- **What you can upload:** one event-pack `.zip` containing its `manifest.json`, **or** one or more loose **PDF, TXT or DOCX** documents. Drag files onto the box or click *choose files*. A zip must be uploaded on its own.
- **Metadata:** each document's header block (DOC_ID, EVENT_ID, TITLE, PUB_DATE, RELIABILITY and so on) supplies its metadata.
- **Result:** the dialog reports how many documents and trade tables were added, how many documents were damaged, and how many suspicious spans (prompt-injection attempts) were flagged. Flagged text is kept as evidence but never followed as an instruction.
- **Who sees it:** questions asked *after* the upload can use the new documents. A case that is already open keeps the evidence it used.

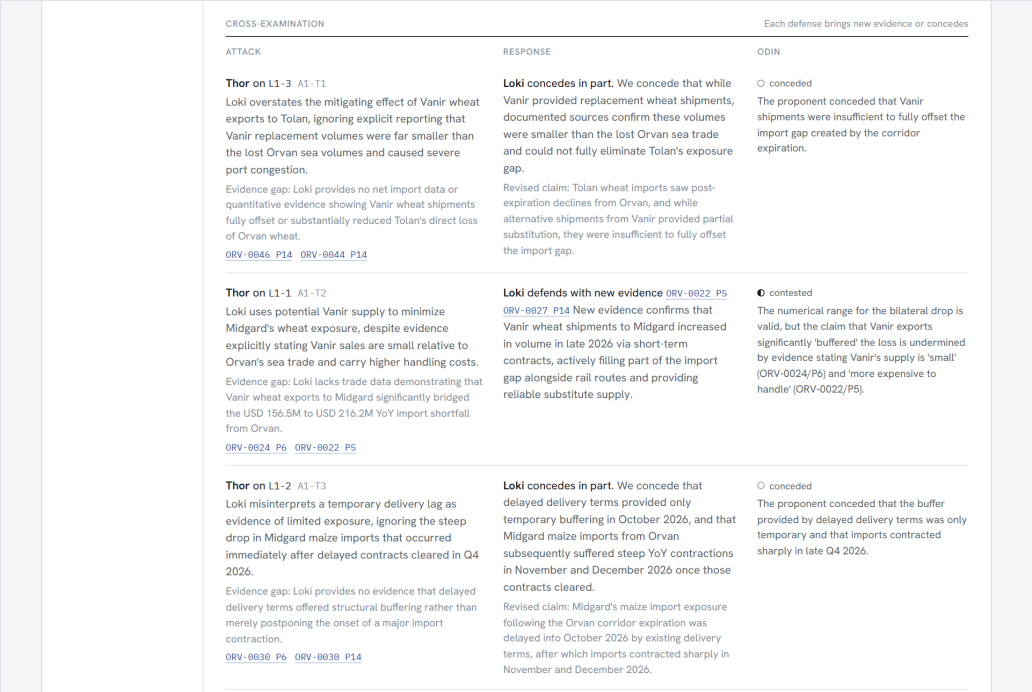
> Uploaded files are validated (type and size), stored under a sanitised name, and parsed as data only. Nothing in a document can change an agent's role, tools, the question, the as-of date or the trial flow.

6. Trial view

The heart of the site: what Heimdall found, what Thor and Loki claimed, how they attacked each other, and how Odin ruled.

From top to bottom

SECTION	WHAT IT SHOWS
Question	The question exactly as asked.
Case file	Heimdall's investigation: **Event** status (established / disproved / uncertain) with the reason and citations; countries and sectors; related countries; routes; trade tables found; key documents (click to open); corrections and rumours; missing data. After a send-back it says *updated for run 2*.
Trial notes	Plain notes the system recorded, for example why run 2 produced no new evidence, or which side went unchallenged.
Rulings summary	Counts of established, contested, conceded and rejected claims (and any not yet ruled).
Thor | Loki	Two columns of claims. Each claim shows its ID (T1-1, L1-2…), **fact** or **inference**, country / sector, Odin's latest verdict, the claim text, its citations and the calculation records behind any numbers (`CALC-…`). Greyed claims were withdrawn or replaced by a revised claim in a later run; a note says which.
Cross-examination	One row per attack, in three columns: **Attack** (who attacked which claim, the objection, the evidence gap and counter-evidence), **Response** (a defense with *new* evidence, a full or partial concession, or "no new evidence, claim stays undefended"), and **Odin** (the verdict and his rationale).

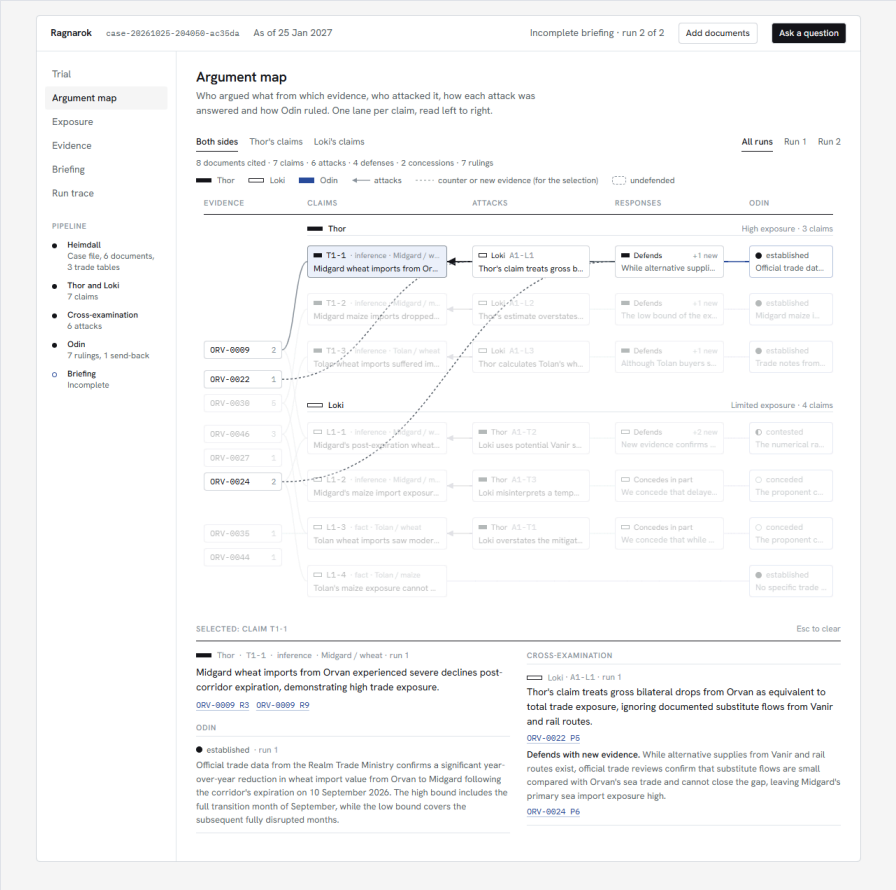


Cross-examination: attack, response, Odin's ruling.

> **Rules the code enforces:** each attack must target a claim and bring counter-evidence or name an evidence gap; a defense counts only if it brings evidence not already cited, otherwise it is shown as undefended; a claim can only be "conceded" if its author conceded it.

7. Argument map

The whole debate as one picture: what each claim rests on, who attacked it, how it was answered and how Odin ruled. It is built from the same recorded events as the Trial view, laid out so the interaction between the agents is visible at a glance.

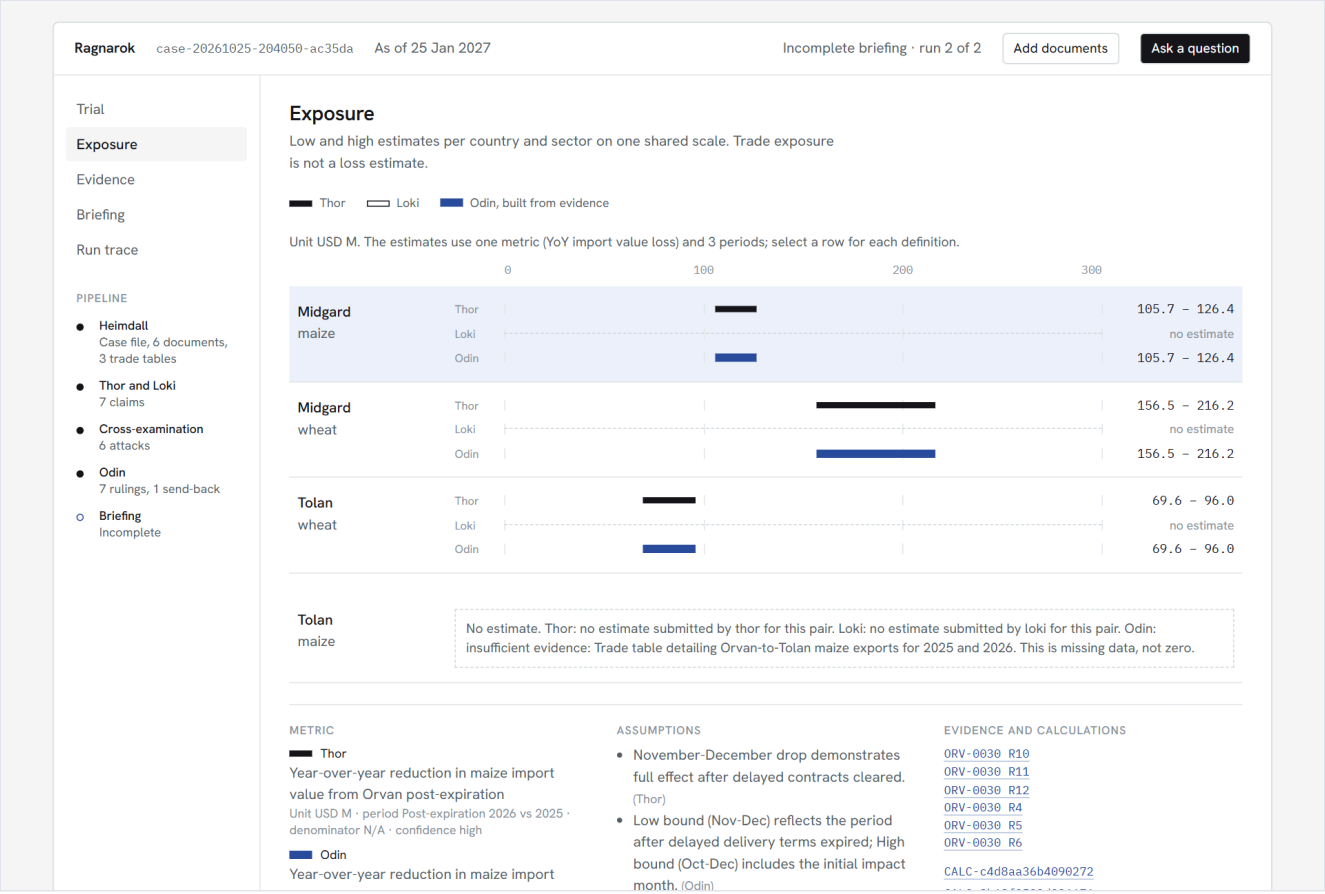


Argument map with claim T1-1 selected: its lane is lit, the rest fades, and the detail panel shows the whole exchange.

PART	WHAT IT SHOWS AND HOW TO USE IT
Columns	Read left to right: **Evidence** (cited documents) → **Claims** → **Attacks** → **Responses** → **Odin**. One lane per claim; Thor's claims in the top band, Loki's below. A revised claim sits directly under the claim it revises.
Marks	Agent bars as everywhere (solid Thor, outlined Loki, blue Odin); Odin's node shows the verdict glyph; a rejected ruling has a red border; an undefended attack has a dashed response box. The small number on a document is how many of its paragraphs or rows were cited.
Hover or select	Hover any node to light up its lane and dim the rest; click to keep it selected (click again, click the background or press Esc to clear). Counter-evidence and new-evidence links (dashed) appear only for the selection, so the map stays readable.
Detail panel	For a claim, attack, response or ruling: the claim with its citations, every attack with its response and new evidence, and Odin's rulings with his rationale. For a document: every use of it ("Thor cites it in T1-1", "Loki uses it against T1-2") with the quote. Every citation opens the Evidence view.
Filters	**Both sides · Thor's claims · Loki's claims**, and **All runs · Run 1 · Run 2** when the case has two runs. A summary line counts documents, claims, attacks, defenses, concessions and rulings.

On a phone the map scrolls sideways inside its own box; the page itself does not.

8. Exposure view

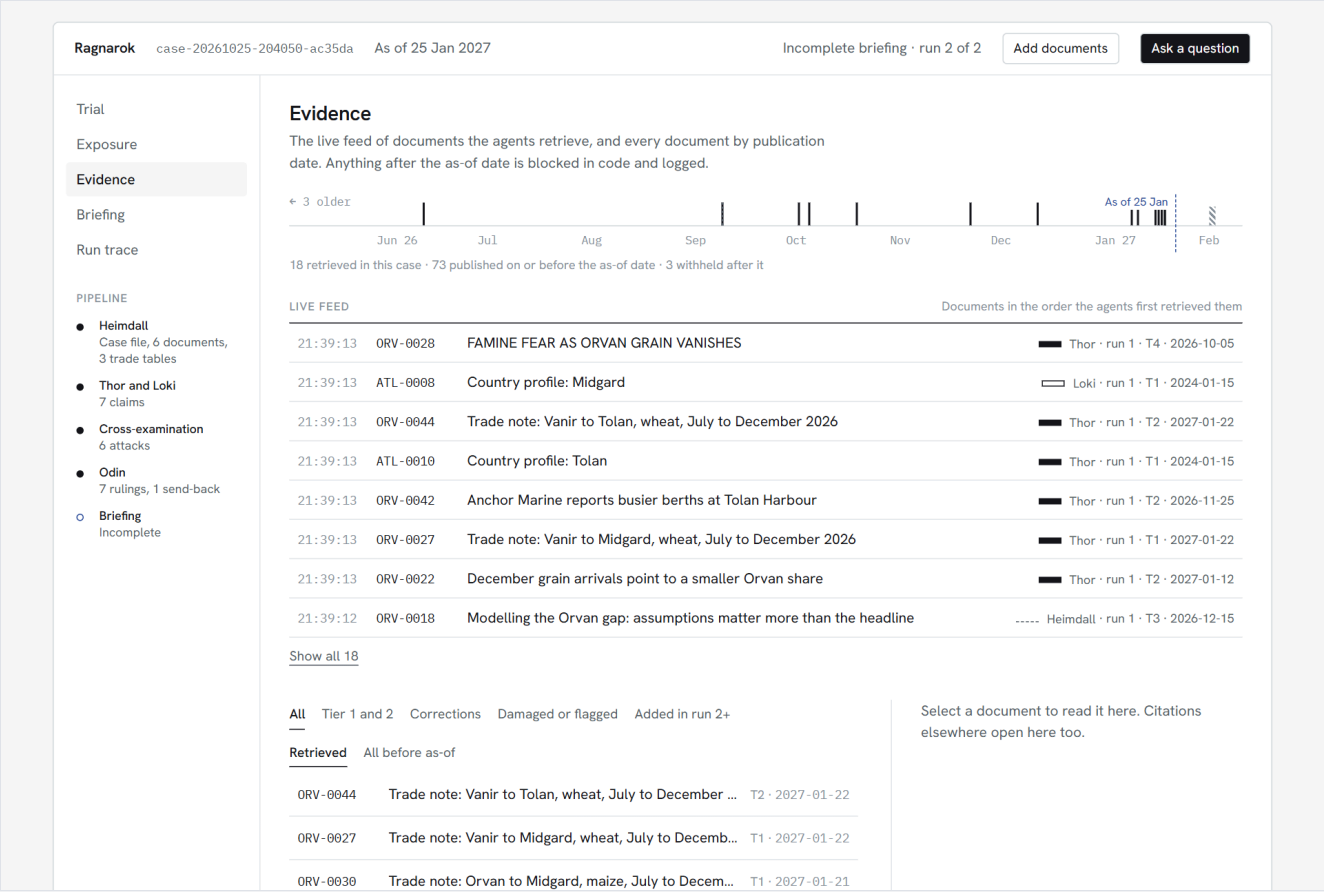


Exposure: one row per country and sector, with Thor's, Loki's and Odin's ranges on one shared axis.

- **Legend:** ▬ Thor (solid) ▭ Loki (outlined) ▬ Odin, built from evidence (blue). Each bar runs from the low to the high bound; the numbers are on the right.
- **Headline** above the chart: the metric, unit and period, read from the estimates themselves. Values in different units are never drawn on the same axis.
- **No estimate** rows have a dashed box with the recorded reason (for example "insufficient evidence: trade table … missing"). **Missing data is never shown as zero.**
- **Click a row** to fill the detail panel below it:
  - **Metric:** each agent's metric definition, unit, period, denominator (when there is one) and confidence.
  - **Assumptions** and **Missing data**, labelled by agent.
  - **Evidence and calculations:** the trade rows used (click to open the row in its table) and the calculation records. Click a `CALC-…` ID to see its operation, result, formula, input rows and any excluded rows with the reason.
- **While a case runs**, the view shows Thor's and Loki's submitted estimates live; Odin's ranges appear with the briefing.

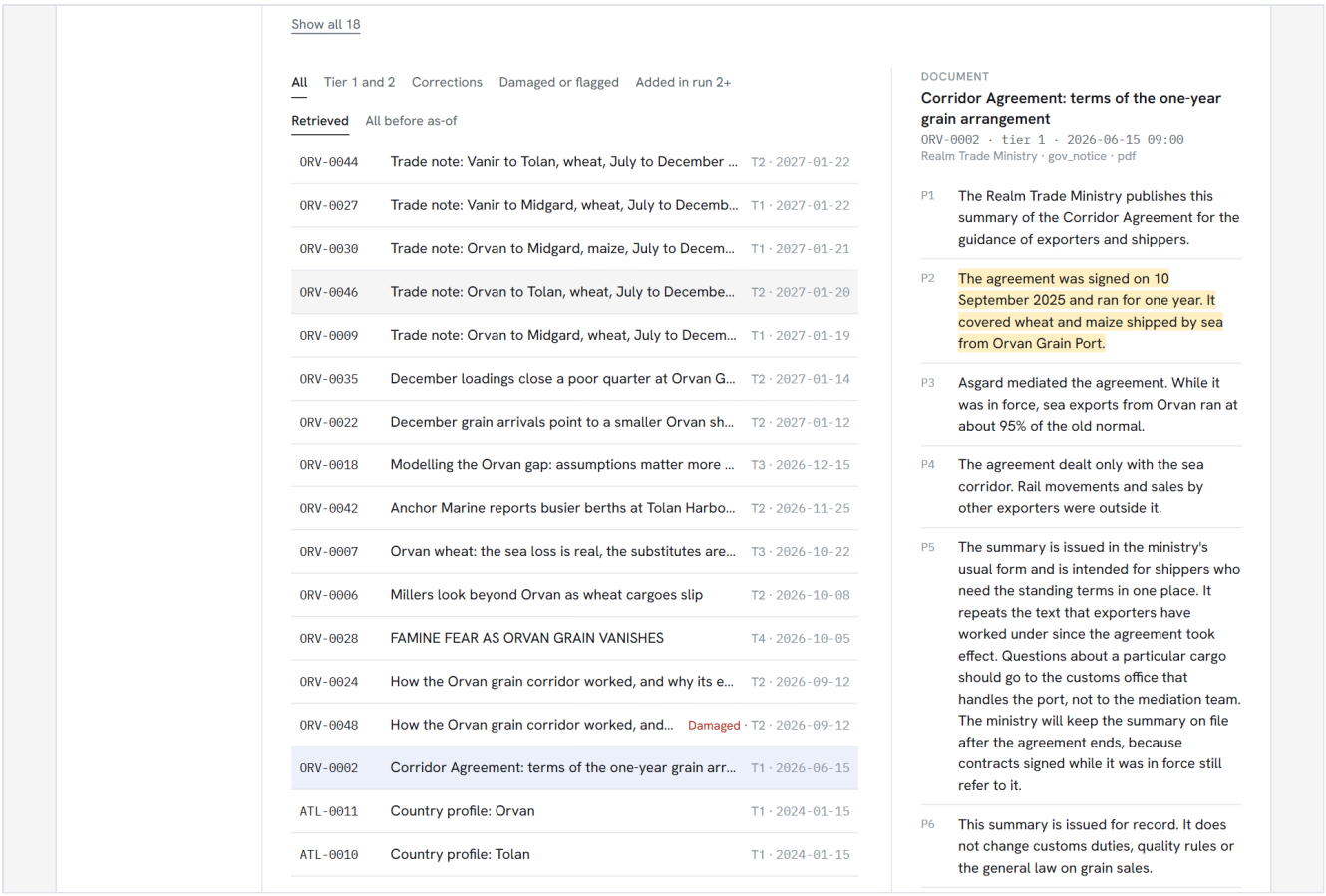
> Trade exposure is the value of trade at risk on the disrupted route. It is not a loss estimate. All arithmetic is done by code from the cited rows, never by the AI model.

9. Evidence view



Evidence: timeline, counts and the live feed.

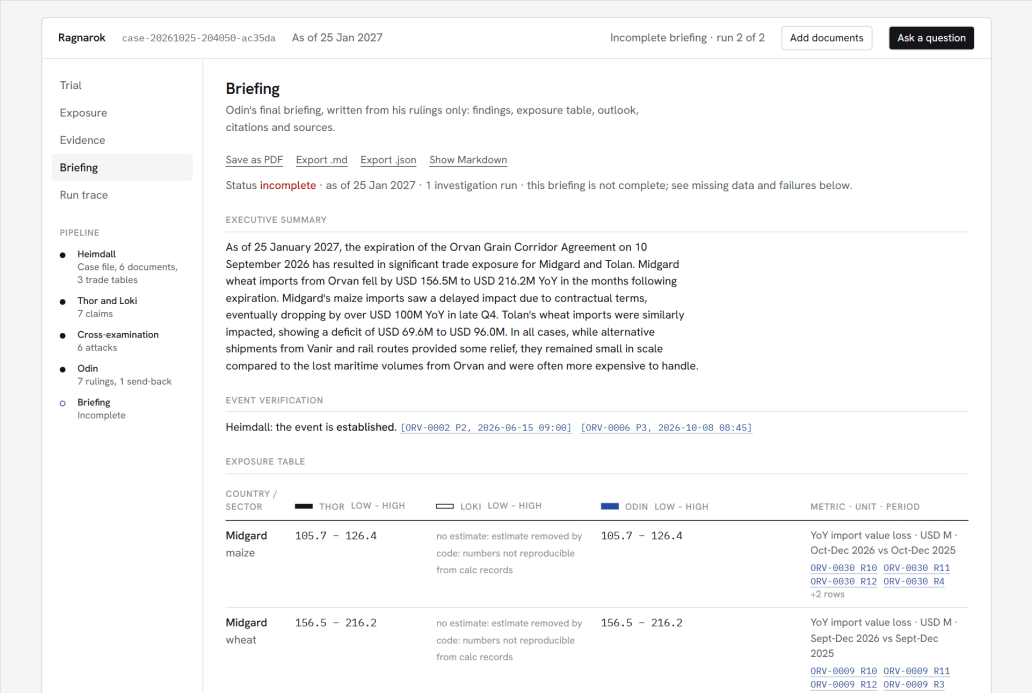
PART	WHAT IT SHOWS AND HOW TO USE IT
Timeline	Every document as a tick by publication date, up to the as-of line. Ticks for documents retrieved in this case are solid; the others are faded. Documents after the as-of date are withheld (shown only as a count). Click a tick to open the document.
Counts line	How many documents were retrieved in this case, how many are published on or before the as-of date, how many are withheld after it, and how many blocked requests were logged.
Live feed	Documents in the order the agents first retrieved them (newest first), with the time, the agent who retrieved them, the run, the tier and the date. Click one to open it at the retrieved paragraph or row. While a case runs it fills in live. Shows the latest 8; *Show all* expands it.
Filters	**All · Tier 1 and 2 · Corrections · Damaged or flagged · Added in run 2+**, and the scope **Retrieved** (only documents this case used) or **All before as-of** (everything the case was allowed to see). "Damaged or flagged" also lists blocked requests for documents published after the as-of date.
Document list	ID, title, reliability tier (T1 most reliable, T4 least), date, and a tag: *Flagged*, *Damaged*, *Corrected by …*, *Corrects …*, *Duplicate of …*, *New in run 2*.
Document reader	The selected document on the right: title, ID, tier, date and time, source, format, its relations (corrections, duplicates), damage notes, and the text paragraph by paragraph (P1, P2…). Cited paragraphs are highlighted in yellow. Flagged paragraphs are tinted red with the note "Suspicious span. Kept as evidence text, withheld as instruction"; a button shows their original text as ingested. For a trade-row citation, the table appears under the text with the cited row highlighted and each row's status (valid, or the damage reason).



Click-through: clicking the citation `ORV-0002 P2` on the Trial view opened this document with paragraph P2 highlighted.

> **The as-of date is enforced by the server, not just hidden in the page.** A document published after the case's as-of date is refused by the API (HTTP 403) and the attempt is logged as `CLOCK_VIOLATION_BLOCKED`. The reader then says the document is outside the case's window.

10. Briefing view



Briefing: export buttons, then the report written from Odin's rulings.

Buttons

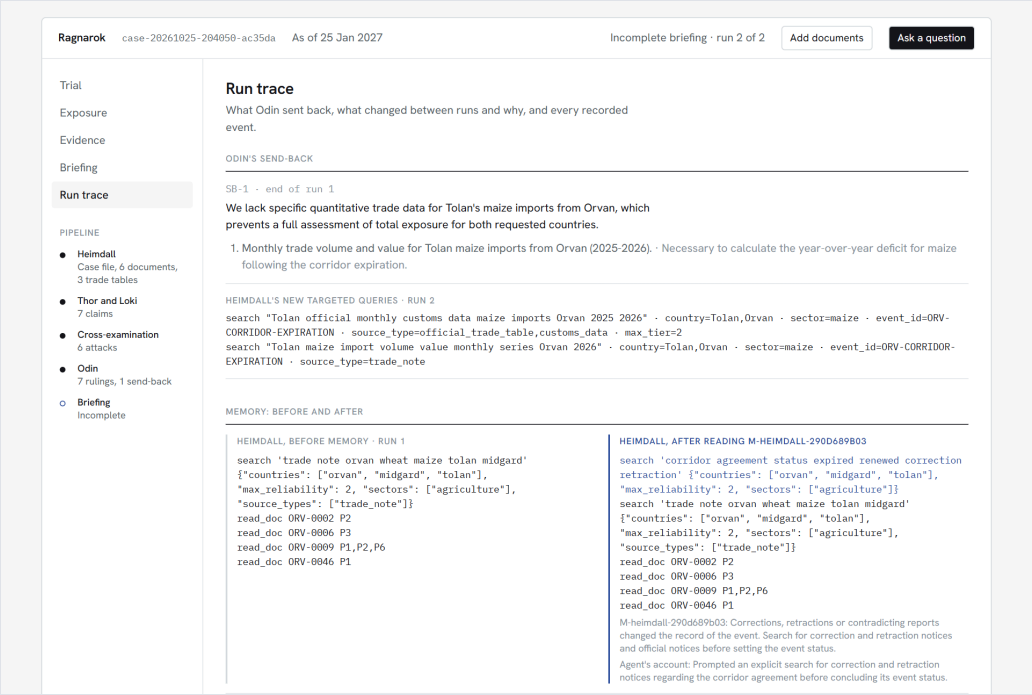
Save as PDF	Opens the browser's print dialog with a clean print layout (no menus). Choose "Save as PDF".
Export .md	Downloads the briefing as Markdown.
Export .json	Downloads the full structured briefing (every finding, estimate and citation).
Show Markdown	Shows the raw Markdown in the page instead of the formatted report.

Sections of the report

1. **Executive summary** and the **briefing status**: *complete*, *partial* or *incomplete* (the site never claims completeness falsely).
2. **Event verification** with citations.
3. **Exposure table:** Thor's, Loki's and Odin's low–high ranges per country and sector, with unit, metric, period and the reason for any missing estimate.
4. **Established, Contested and Conceded findings**, each marked fact or inference, with inline citations in the form `[ORV-0009 R3, 2027-01-19 09:00]`: document and paragraph or trade row, plus publication date and time.
5. **Rejected claims**, **Next-quarter outlook** with its explicit assumptions, **Assumptions and uncertainty**, **Missing data and evidence quality**, and **Traps, corrections and failed operations** (injection attempts, blocked documents, damaged rows, failures).
6. **Sources:** every document cited, labelled as Runemark sources.

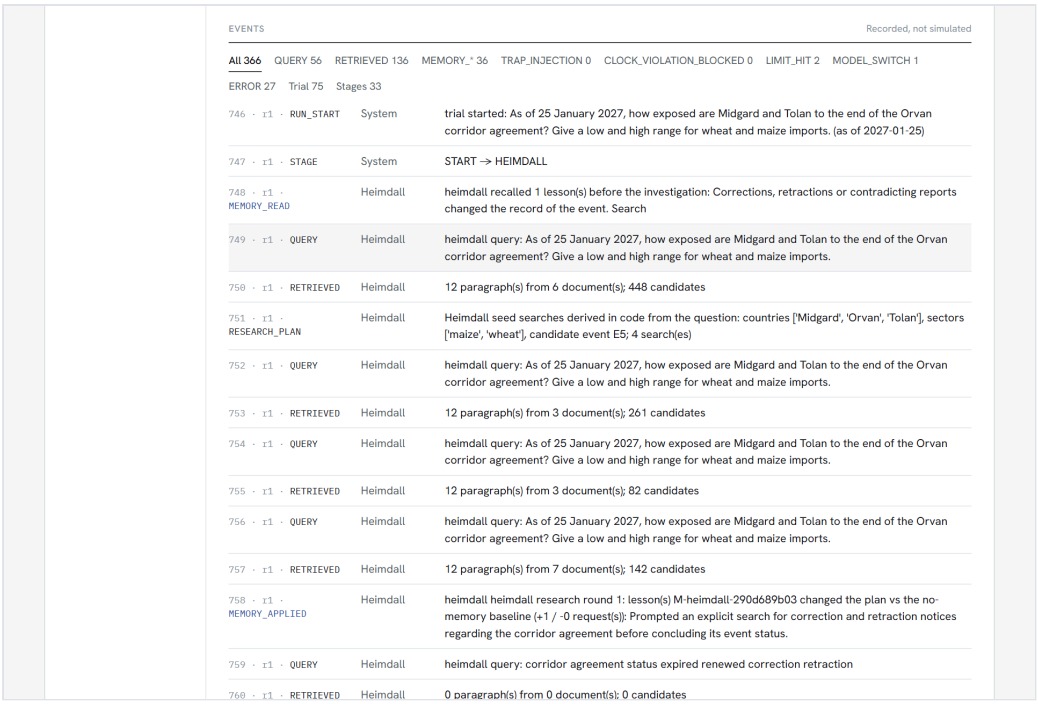
Every citation is checked when the briefing is built (the document and paragraph or row exist, the quote matches, the date is within the as-of window). Anything that fails is dropped and listed under "Dropped at build time". A case that failed or is paused has no briefing yet; the view says so.

11. Run trace view



Run trace: send-back, memory before/after, run comparison.

SECTION	WHAT IT SHOWS
Odin's send-back	Why Odin sent the case back: what is missing, why it matters and which claims it affects. Below it, **Heimdall's new targeted queries** for run 2, with their filters.
Memory: before and after	Each agent has a private memory that lasts between cases. When reading a memory changed what an agent did, this shows the planned action **before** memory and the changed action **after**, with the memory IDs used.
Run 1 against run 2	A side-by-side table: evidence retrieved, genuinely new evidence in run 2, claims, withdrawals and revisions, verdict changes, Odin's sufficiency decision, and the briefing status. A note summarises the effect of the new evidence.
Events	The complete append-only audit log, labelled "Recorded, not simulated". Filter by type: **QUERY, RETRIEVED, MEMORY_*, TRAP_INJECTION, CLOCK_VIOLATION_BLOCKED, LIMIT_HIT, MODEL_SWITCH** and more, each with its count. Click any event to expand its full recorded details. Shows 60 at a time; a button loads more.



The event log with type filters.

12. Marks, colours and labels

Verdicts (Odin's rulings)

MARK	VERDICT	MEANING
●	established	Supported by valid, timely, relevant evidence that survived cross-examination.
◐	contested	Some support, but credible counter-evidence or an open gap remains.
○	conceded	The claim's own author withdrew it or acknowledged the objection.
■	rejected	Odin judged it unsupported or invalid (bad citation, wrong reading, non-reproducible number).
◌	not ruled	No ruling recorded yet (or the case stopped first).

Conceded and rejected are different: a concession comes from the advocate, a rejection from Odin.

Agents

Thor (solid bar) Loki (outlined bar) Odin (blue bar)

Other labels

ORV-0002 P2	A citation: document ID plus paragraph (P…) or trade row (R…). Always clickable.
CALC-cb98e8	A deterministic calculation record (sum, share, year-on-year change) with its input rows.
fact / inference	A fact is stated directly in a source; an inference is the agent's reasoning from sources.
T1 … T4	Outlet credibility tier from the pack's reliability model: T1 is the most reliable (for example official notices), T4 the least (rumour, social media, tabloids).
Flagged	The document contains a suspicious span (for example a fake "SYSTEM:" instruction). Kept as data, never obeyed.
Damaged	The file or a table row is cut off, garbled or unreadable. Kept for provenance; damaged rows are never used as valid numbers.
Corrected by / Corrects	A later document corrects this one (or this one corrects another). Both are kept, linked.
Duplicate of	Same body text as another document; excluded from normal search, provenance kept.
highlight	The paragraph or row that a citation points to.

13. Case statuses and alerts

TOP-BAR STATUS	MEANING	WHAT TO DO
Running · <stage> · run N of M	The trial is in progress. Views fill in live from recorded events.	Wait, or browse the views.
Paused · daily model quota · resumes after <time>	Every configured AI model has used its daily request quota. A yellow alert explains it. Nothing is lost.	Leave the API running; the case resumes on its own after the reset time. Finished steps replay from the cache without new requests. It can also be resumed with `POST /cases/{id}/resume`.
Complete briefing	Odin judged the evidence sufficient.	Read and export the briefing.
Partial / Incomplete briefing	A briefing exists but evidence was still missing after the permitted runs, or a limit stopped the trial. It is labelled as such.	Read the missing-data notes; add documents and ask again if you have more evidence.
Stopped: event disproved	Heimdall found the event did not happen, so there was nothing to argue.	Check the case file's event reasons.
Failed at <stage> · no briefing	An unrecoverable error. A red alert lists the last notes and a limitation report; partial results recorded before the failure are still shown.	Read the limitation report; resume the case via the API.
Interrupted at <stage>	The API stopped while the case was running.	Resume the case via the API.

Other alerts

- **The trial stopped:** a hard limit (runs, tool calls, tokens, evidence volume) was reached; the reason is given.
- **Live status unavailable:** the live connection to the API dropped; the page keeps showing what was recorded.
- **Case unavailable:** the case ID does not exist or the API is not running.
- **Added N document(s)…:** on the Evidence view after an upload.

Loading states show grey placeholder rows; empty states say plainly what is missing ("No claims yet. They appear here as Thor and Loki submit them.").

14. Common tasks

I WANT TO...	DO THIS
Start a new investigation	Ask a question → type the question → pick the as-of date → Start investigation.
Reopen a saved case	Ask a question → Open a saved case → paste the ID → Open. Or open its link.
Verify a claim's source	Trial → click the claim's citation → Evidence opens with the paragraph highlighted.
See how a number was computed	Exposure → click the row → click a `CALC-...` ID under "Evidence and calculations".
See the whole debate at a glance	Argument map → hover or select a claim or document.
Compare Thor, Loki and Odin	Exposure view (chart) or Briefing → Exposure table.
See why a case went to a second run	Run trace → Odin's send-back, then Run 1 against run 2.
See memory changing an agent's behaviour	Run trace → Memory: before and after.
See prompt-injection attempts	Evidence → filter "Damaged or flagged"; or Run trace → Events → TRAP_INJECTION.
See blocked future documents	Run trace → Events → CLOCK_VIOLATION_BLOCKED; or Evidence → "Damaged or flagged".
Add a new event pack	Add documents → drop the pack .zip (with manifest.json) → Add pack. Then ask a question about it.
Add one or two documents	Add documents → drop the PDF / TXT / DOCX files → Add documents.
Get the report as a file	Briefing → Save as PDF, Export .md or Export .json.

15. Troubleshooting

PROBLEM	CAUSE AND FIX
Page is blank or says "Case unavailable"	The API on port 8000 is not running, or is an old process. Start (or restart) it with the command in section 2.
Start investigation is greyed out	The question is shorter than 5 characters, no as-of date is picked, or another case is running.
Case shows "Paused"	The free-tier daily model quota is used up. Keep the API running; it resumes after the time shown.
A cited document says it is outside the window	It was published after the case's as-of date; the server refuses it on purpose.
Upload says "Not a supported file"	Only .zip (pack), .pdf, .txt and .docx are accepted; a zip must be uploaded alone.
An exposure row says "No estimate"	The evidence did not support a range; the reason is shown. This is missing data, not zero.

Run 2 shows no new evidence	Heimdall's targeted searches found nothing beyond run 1; the trial notes say so and the briefing is labelled incomplete.

16. Component map (for developers)

The dashboard is React + Vite + TypeScript + Tailwind, in `frontend/src/`. It only reads the API and the recorded event stream.

FILE	RESPONSIBILITY
`App.tsx`	Page frame; picks the view from the URL; start page; failed, paused, stopped and upload alerts; turns any citation click into "open the Evidence view at this paragraph or row".
`useCase.ts`	Loads the case status (`GET /cases/{id}`), follows the live event stream (SSE, with a polling fallback) and derives the trial state: claims, attacks, defenses, concessions, rulings, send-backs, memory events, live feed, calculations.
`api.ts`	Typed API client and shared data types.
`components/Shell.tsx`	Top bar (case ID, as-of, run status, Add documents, Ask a question) and side rail (views and pipeline).
`components/CaseForm.tsx`	Question + as-of form and "Open a saved case".
`components/UploadDialog.tsx`	Upload form for a pack zip or loose documents (`POST /ingest`).
`components/marks.tsx`	Shared marks: verdict glyphs, agent bars, fact/inference label, clickable citation.
`components/layout.tsx`	Section, heading, alert, empty state, loading skeleton, filter tabs.
`components/ui/dialog.tsx`	Accessible modal dialog (Radix).
`views/ArgumentMapView.tsx · lib/argmap.ts`	Argument map: lane layout (pure function) and the interactive graph with its detail panel.
`views/TrialView.tsx`	Case file, trial notes, rulings summary, Thor/Loki claims, cross-examination.
`views/ExposureView.tsx`	Range chart per country and sector, detail panel, calculation records.
`views/EvidenceView.tsx`	Timeline, live feed, filters, document list, document reader and trade-table rows (`GET /feed`, `/docs`, `/trade`).
`views/BriefingView.tsx`	Rendered briefing, PDF print, Markdown and JSON export.
`views/TraceView.tsx`	Send-back, memory before/after, run comparison, filtered event log.
`lib/route.ts · format.ts · scale.ts`	URL routing (`#case=…&view=…`), date and number formatting, chart scale.

API endpoints the website uses

ENDPOINT	USED FOR
`POST /cases`	Start a case (one at a time).
`GET /cases/{id}`	Status (running / complete / partial / failed / paused) and results.
`GET /cases/{id}/events`	Live Server-Sent Events of the recorded audit log.
`POST /cases/{id}/resume`	Resume a failed, partial or paused case.
`GET /docs/{id}, GET /trade/{table}`	Documents and trade rows, gated by the case's as-of date.
`GET /feed?as_of=`	Documents up to the as-of date with tier, correction, damage and injection badges.
`POST /ingest`	Add a pack zip or loose PDF / TXT / DOCX files.